

User and Group

These two control which user and group Apache runs as. This is very important for the security of the server. If you tell Apache to run as the root user and group, you are giving the Web server the power of the systems administrator! It makes no sense to do that, unless you are a developer and are testing something. Never do this.

The ideal value for user and group is `www:www`. With this, you can control which files Apache can write to, which files it has read-only access to, and so on, without ever needing to do a `chmod 777`!

Example:

```
User www
```

```
Group www
```

Timeout

This is the time period that Apache should wait before dropping a request due to a very long *send* or *receive* operation.

For a more precise definition (taken from the Apache manual), *Timeout* controls the following:

- The total time Apache takes to receive a GET request.
- The amount of time between the receipts of TCP packets on a POST or PUT requests.
- The amount of time between ACKs on transmissions of TCP packets, in responses.

The default value for *Timeout* is 300 seconds (specified as *Timeout 300*). At the time of writing, there's no way to configure separate time-outs for each element mentioned above.

KeepAlive

This option specifies if Apache should allow multiple requests per connection (i.e., persistent connections, when *KeepAlive On* is specified). This should be kept on for performance reasons; otherwise, every time a new request arrives (multiple requests from the same client), Apache creates a new connection, which has a lot of overhead.

As a simple analogy, consider a group of 100 people waiting to enter through the door of some institution. If the watchman closes and opens the door once for each person, it would take much more time and effort, compared to letting all 100 people enter at one time.

MaxKeepAliveRequests

This option specifies the maximum number of requests that will be permitted on a single connection (if enabled using *KeepAlive*), before Apache closes the connection and initiates a new one for a new request from the same client. This number is better left high, for performance reasons. The default value is 0, which means 'unlimited'. For example, you could specify *MaxKeepAliveRequests 1000*.

KeepAliveTimeout

This option specifies the maximum time (in seconds) that Apache should wait for a subsequent request from the same client, on an existing persistent connection, before freeing up the process for another client. Continuing with our simple analogy, this is the maximum time the watchman should keep the gate open, waiting for the arrival of another person who also wishes to enter.

This value should be between 5 and 15. Setting it to less than 5 makes no sense, since a browser could easily request another file from the server within 5 seconds; setting it higher than 15 seconds would unnecessarily tie up an Apache process, which could be used in handling another client.

Example:

```
KeepAliveTimeout 5
```

AccessFileName

This option specifies the name of the file that must be read, if present in a directory, to apply directory-specific options. As mentioned earlier, the name is usually *.htaccess* (i.e., *AccessFileName .htaccess*). There are performance considerations with the use of these configuration files. If you have set any of the options permitted by the *AllowOverride* directive (see below) in the relevant *<Directory>* section, Apache will check for *.htaccess* files in every directory from the top-most specified in *<Directory>* up to the last directory (in which the requested file resides). This causes a lot of overhead, and is not recommended for a server receiving a large number of requests per second. A workaround, if you really want the *.htaccess* feature (as is the case with shared hosts), is to have a specific *<Directory>* section where all your documents reside, instead of enabling that option in the parent directory.

AllowOverride

This directive lets you permit certain configuration options in files of the name given in *AccessFileName*, according to the values specified in this directive. It can appear only in *<Directory>* sections specified without regular expressions (not containing ~), and nowhere else.

The values it takes, and what it does, is given in Table 1, which is taken from the official Apache documentation at <http://j.mp/h2lVUU>.

HostnameLookups

Enabling this option gives you host-names in the access log, instead of just IP addresses. This is a big performance hit, since every time a client initiates a connection, Apache will send a request to the DNS server to convert the IP address of the client to a host-name. Disable this (*HostnameLookups Off*). If you want host-names in the log file, use the *logresolve* utility provided with Apache instead, to resolve IP addresses when you are reading the log.